
Berkovich Space

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1 Spectrum

Let $\mathbf{k}$ be a spherically complete non-archimedean field which is algebraically closed and $A = \mathbf{k}[T]$. We want to consider the “analytic structure” on $\mathbf{mSpec} A$. However, unlike the complex case, the set $\mathbf{mSpec} A$ is totally disconnected with respect to the topology induced by the absolute value on $\mathbf{k}$ (??). To overcome this difficulty, Berkovich uses multiplicative semi-norms to “fill in the gaps” between the points in $\mathbf{mSpec} A$, leading to the notion of the spectrum of a Banach ring.

1.1 Definition and examples

Definition 1.1. Let R be a Banach ring. The *Berkovich spectrum* $\mathcal{M}(R)$ of R is defined as the set of all multiplicative semi-norms on R that are bounded with respect to the given norm on R . For every point $x \in \mathcal{M}(R)$, we denote the corresponding multiplicative semi-norm by $|\cdot|_x$.

We equip $\mathcal{M}(R)$ with the weakest topology such that for each $f \in R$, the evaluation map $\mathcal{M}(R) \rightarrow \mathbb{R}_{\geq 0}$, defined by $x \mapsto |f|_x =: f(x)$, is continuous.

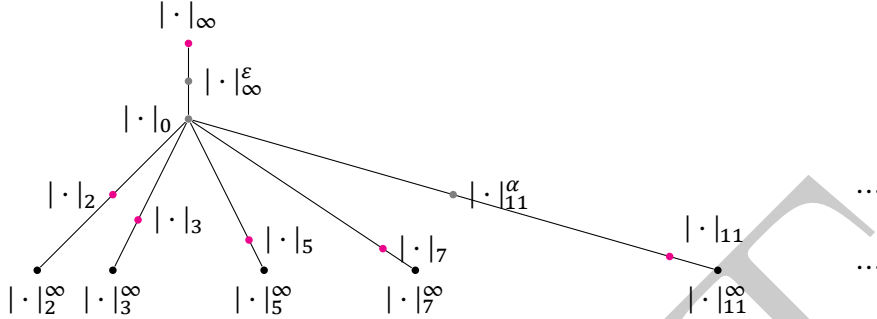
Example 1.2. Let $(\mathbf{k}, |\cdot|)$ be a complete valuation field. The Berkovich spectrum $\mathcal{M}(\mathbf{k})$ consists of a single point corresponding to the given absolute value $|\cdot|$ on $\mathbf{k}$.

Example 1.3. Consider the Banach ring $(\mathbb{Z}, \|\cdot\|)$ with $\|\cdot\| = |\cdot|_{\infty}$ is the usual absolute value norm on $\mathbb{Z}$. Let $|\cdot|_p$ denote the p -adic norm for each prime number p , i.e., $|n|_p = p^{-v_p(n)}$ for each $n \in \mathbb{Z}$,

where $v_p(n)$ is the p -adic valuation of n . The Berkovich spectrum

$$\mathcal{M}(\mathbb{Z}) = \{|\cdot|_\infty^\varepsilon : \varepsilon \in (0, 1]\} \cup \{|\cdot|_p^\alpha : p \text{ is prime}, \alpha \in (0, \infty]\} \cup \{|\cdot|_0\},$$

where $|a|_p^\infty := \lim_{\alpha \rightarrow \infty} |a|_p^\alpha$ for each $a \in \mathbb{Z}$ and $|\cdot|_0$ is the trivial norm on $\mathbb{Z}$.



Yang: To be continued.

Theorem 1.4. Let R be a Banach ring. The spectrum $\mathcal{M}(R)$ is nonempty.

Proof. Let $\mathfrak{m}$ be a maximal ideal of R . Note that the pullback of residue norm on the residue field $R/\mathfrak{m}$ is bounded with respect to the given norm on R . Replacing R by the completion of $R/\mathfrak{m}$, we may assume that R is a complete field. Consider the set

$$\Sigma = \{\text{norm on } R \text{ bounded by the given norm } \|\cdot\|\}$$

with the partial order defined by boundedness. Since for a descending chain in Σ , the infimum is a norm, by Zorn's lemma, there exists a minimal element $|\cdot| \in \Sigma$.

We claim that $|\cdot|$ is multiplicative. Since the spectral radius $\rho(f) = \lim_{n \rightarrow \infty} |f^n|^{1/n}$ associated to $|\cdot|$ is power-multiplicative and bounded by $|\cdot|$, by minimality of $|\cdot|$, we have $\rho(f) = |f|$ for each $f \in R$. Thus $|\cdot|$ is power-multiplicative. If $|\cdot|$ is not multiplicative, then there exist $a, b \in R \setminus \{0\}$ such that $|ab| < |a||b|$. Then $|b| \leq |a^{-1}||ab| < |a^{-1}||a||b|$, which implies that $|a||a^{-1}| > 1$. Set $r = |a|^{-1} < |a^{-1}|$ and consider $R\langle T/r \rangle$. Since $r \cdot |a| = 1$, we have that

$$\left| \sum_{n=0}^{\infty} a^n T^n \right| = \sum_{n=0}^{\infty} |a^n| r^n = \sum_{n=0}^{\infty} |a|^n r^n = \sum_{n=0}^{\infty} 1 = \infty.$$

The power series is not convergent in $R\langle T/r \rangle$ and hence $1 - aT$ is not invertible in $R\langle T/r \rangle$. Let $\mathfrak{n}$ be a maximal ideal of $R\langle T/r \rangle$ containing $1 - aT$. Consider $R \rightarrow R\langle T/r \rangle \rightarrow R\langle T/r \rangle/\mathfrak{n}$. Since R is a field, the composition is injective. The residue norm on $R\langle T/r \rangle/\mathfrak{n}$ induces a norm $|\cdot|'$ on R bounded by $|\cdot|$. Note that $|a^{-1}'| \leq |T| = r = |a|^{-1} < |a^{-1}|$, contradicting the minimality of $|\cdot|$. $\square$

Definition 1.5. Let $\varphi : R \rightarrow S$ be a bounded ring homomorphism of Banach rings. The pullback map $\mathcal{M}(\varphi) : \mathcal{M}(S) \rightarrow \mathcal{M}(R)$ is defined by $\mathcal{M}(\varphi)(x) = x \circ \varphi : f \mapsto |\varphi(f)|_x$ for each $x \in \mathcal{M}(S)$.

Note that $\mathcal{M}(\varphi)(f^{-1}(V)) = \varphi(f)^{-1}(V)$ for each $f \in R$ and open subset $V \subset \mathbb{R}_{\geq 0}$. Hence the pullback map is continuous.

Notation 1.6. Let R be a Banach ring and $x \in \mathcal{M}(R)$. We denote by $|\cdot|_x$ the multiplicative semi-norm on R corresponding to the point x . Its kernel $\{f \in R : |f|_x = 0\}$ is a closed prime ideal of R , denoted by $\wp_x$.

Definition 1.7. Let R be a Banach ring. For each $x \in \mathcal{M}(R)$, the *completed residue field* at the point x is defined as the completion of the residue field $\kappa(x) = \text{Frac}(R/\wp_x)$ with respect to the multiplicative norm induced by the semi-norm $|\cdot|_x$, denoted by $\mathcal{H}(x)$.

Example 1.8. Consider the Banach ring $(\mathbb{Z}, |\cdot|_\infty)$ as in Example 1.3. We have

- $x = |\cdot|_\infty^\varepsilon$ for some $\varepsilon \in (0, 1]$: $\wp_x = (0)$ and $\mathcal{H}(x) \cong \mathbb{R}$ with the absolute value norm raised to the power ε ;
- $x = |\cdot|_0$: $\wp_x = (0)$ and $\mathcal{H}(x) \cong \mathbb{Q}$ with the trivial norm;
- $x = |\cdot|_p^\alpha$ for some prime number p and $\alpha \in (0, \infty)$: $\wp_x = (0)$ and $\mathcal{H}(x) \cong \mathbb{Q}_p$ with the p -adic norm raised to the power α ;
- $x = |\cdot|_p^\infty$ for some prime number p : $\wp_x = (p)$ and $\mathcal{H}(x) \cong \mathbb{F}_p$ with the trivial norm.

Definition 1.9. Let R be a Banach ring. The *Gel'fand transform* of R is the bounded ring homomorphism

$$\Gamma : R \rightarrow \prod_{x \in \mathcal{M}(R)} \mathcal{H}(x), \quad f \mapsto (f(x))_{x \in \mathcal{M}(R)},$$

where the norm on the product $\prod_{x \in \mathcal{M}(R)} \mathcal{H}(x)$ is given by the supremum norm.

Example 1.10. Let K be a number field equipped with Yang: To be continued.

Proposition 1.11. The Gel'fand transform $\Gamma : R \rightarrow \prod_{x \in \mathcal{M}(R)} \mathcal{H}(x)$ of a Banach ring R factors through the uniformization R^u of R , and the induced map $R^u \rightarrow \prod_{x \in \mathcal{M}(R)} \mathcal{H}(x)$ is an isometric embedding. Yang: To be checked.

Proof. Yang: To be added. □

Lemma 1.12. Let $\{K_i\}_{i \in I}$ be a family of completed fields. Consider the Banach ring $R = \prod_{i \in I} K_i$ equipped with the product norm. The spectrum $\mathcal{M}(R)$ is homeomorphic to the Stone-Čech compactification of the discrete space I .

Proof. Yang: To be added. □

Remark 1.13. Here we list some facts about ultrafilters and the Stone-Čech compactification that will be used in the proof of Lemma 1.12.

Let I be a set. A *filter* on I is a nonempty collection $\mathcal{F} \subset \mathcal{P}(I)$ of subsets of I such that:

- if $A, B \in \mathcal{F}$, then $A \cap B \in \mathcal{F}$;
- if $A \in \mathcal{F}$ and $A \subset B \subset I$, then $B \in \mathcal{F}$.

For example, let $I = \mathbb{N}$ and

$$\mathcal{F} = \{A \subset \mathbb{N} : \#(\mathbb{N} \setminus A) < \infty\}.$$

Then $\mathcal{F}$ is a filter on $\mathbb{N}$, called the *Fréchet filter*. Another example is the filter called the *principal filter* generated by a point $i_0 \in I$:

$$\mathcal{F}_{i_0} = \{A \subset I : i_0 \in A\}.$$

Filters are partially ordered by inclusion. A filter $\mathcal{U}$ is called an *ultrafilter* if it is maximal with respect to inclusion. A filter is an ultrafilter if and only if for every subset $A \subset I$, either $A \in \mathcal{U}$ or $I \setminus A \in \mathcal{U}$.

One can define the notion of convergence along ultrafilters in a topological space. Let X be a topological space, $(x_i)_{i \in I}$ a collection of points in X indexed by a set I , and $\mathcal{U}$ an ultrafilter on I . We say that $(x_i)_{i \in I}$ *converges* to a point $x \in X$ along the ultrafilter $\mathcal{U}$ if for every open neighborhood U of x , the set $\{i \in I : x_i \in U\}$ belongs to $\mathcal{U}$. It is denoted by $\lim_{\mathcal{U}} x_i = x$. When $I = \mathbb{N}$ and $\mathcal{U}$ is the Fréchet filter, this notion of convergence coincides with the usual notion of convergence of sequences.

The Stone-Čech compactification of a discrete space is the largest compact Hausdorff space in which the original space can be densely embedded. Yang: To be checked.

Theorem 1.14. Let R be a Banach ring. The spectrum $\mathcal{M}(R)$ is a compact Hausdorff space.

Proof. Yang: To be added. □

Corollary 1.15. Let R be a Banach ring and R^u its uniformization. The canonical map $\mathcal{M}(R^u) \rightarrow \mathcal{M}(R)$ induced by the bounded homomorphism $R \rightarrow R^u$ is a homeomorphism.

Proposition 1.16. Let $\mathbf{l}/\mathbf{k}$ be a finite normal extension of complete valuation fields and A a Banach $\mathbf{k}$ -algebra. Then we have:

- (a) if $\mathbf{l}/\mathbf{k}$ is Galois, then $\mathcal{M}(A \otimes_{\mathbf{k}} \mathbf{l})$ admits a natural action of the Galois group $\text{Gal}(\mathbf{l}/\mathbf{k})$ and the quotient space is homeomorphic to $\mathcal{M}(A)$;
- (b) if $\mathbf{l}/\mathbf{k}$ is purely inseparable, then the canonical map $\mathcal{M}(A \otimes_{\mathbf{k}} \mathbf{l}) \rightarrow \mathcal{M}(A)$ is a homeomorphism.

Proof. Yang: To be added. □

Corollary 1.17. Let $\mathbf{k}$ be a complete valuation field, $\mathbb{k}$ its completion of an algebraic closure, and A a Banach $\mathbf{k}$ -algebra. Then we have a natural homeomorphism $\mathcal{M}(A \hat{\otimes}_{\mathbf{k}} \mathbb{k}) / \text{Gal}(\mathbb{k}/\mathbf{k}) \cong \mathcal{M}(A)$.

Proof. Yang: To be added. □

1.2 Reduction map and kernel map

Proposition 1.18. Let R be a Banach ring. The kernel map $\mathcal{M}(R) \rightarrow \text{Spec}(R), x \mapsto \wp_x$ is continuous with respect to the Zariski topology on $\text{Spec}(R)$.

Proof. Let $D(f) = \{f \neq 0\} \subset \text{Spec}(R)$ be a principal open subset for some $f \in R$. The preimage of $D(f)$ under the kernel map is just the set $\{x \in \mathcal{M}(R) : |f|_x > 0\} = f^{-1}(\mathbb{R}_{>0})$, which is open in $\mathcal{M}(R)$ by definition of the topology on $\mathcal{M}(R)$. □

Example 1.19. Let us consider the spectrum $\mathcal{M}(\mathbb{Z})$ in Example 1.3. Under the kernel map $\mathcal{M}(\mathbb{Z}) \rightarrow \text{Spec}(\mathbb{Z})$, the points $|\cdot|_p^\alpha$ for each prime number p are mapped to the prime ideal (p) , the other above points are all mapped to the zero ideal (0) .

Yang: Is this surjective? what is its fiber?

Yang: Seems i really need the surjectivity to analysis spectrum of $\mathbb{k}[T]$.

Proposition 1.20. Let A be an $\mathbf{k}$ -affinoid algebra over a complete non-archimedean field $\mathbf{k}$. Then the kernel map $\mathcal{M}(A) \rightarrow \text{Spec}(A)$ is surjective.

Proof. Yang: To be add. □

Construction 1.21. Suppose that R is a non-archimedean Banach ring with valuation subring R° and maximal ideal $R^{\circ\circ}$. For each $x \in \mathcal{M}(R)$, there is an induced homomorphism $R^\circ \rightarrow \mathcal{H}(x)^\circ$ between the valuation subrings. Furthermore, we have an induced homomorphism between the residue rings $\tilde{R} = R^\circ/R^{\circ\circ} \rightarrow \mathcal{H}(x)$. This gives rise to the *reduction map*

$$\text{Red} : \mathcal{M}(R) \rightarrow \text{Spec}(\tilde{R}), \quad x \mapsto \ker(\tilde{R} \rightarrow \mathcal{H}(x)).$$

Let $f : R \rightarrow S$ be a bounded homomorphism of non-archimedean Banach rings. Yang: The following diagram commutes:

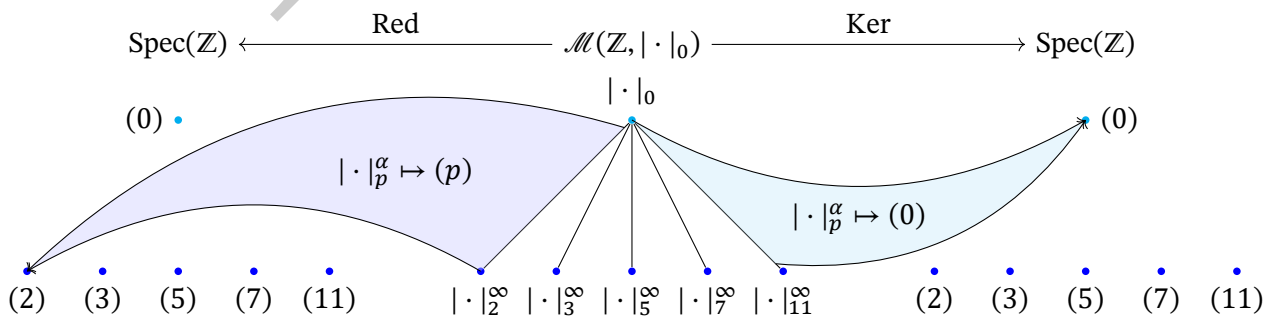
$$\begin{array}{ccc} \mathcal{M}(S) & \xrightarrow{\text{Red}} & \text{Spec}(\tilde{S}) \\ \mathcal{M}(f) \downarrow & & \downarrow \text{Spec}(\tilde{f}) \\ \mathcal{M}(R) & \xrightarrow{\text{Red}} & \text{Spec}(\tilde{R}) \end{array}$$

Hence the reduction map is functorial with respect to bounded homomorphisms of non-archimedean Banach rings.

Example 1.22. Let $(\mathbb{Z}, |\cdot|_0)$ be the Banach ring with the trivial norm. The reduction ring is $\tilde{\mathbb{Z}} = \mathbb{Z}$.

- $x = |\cdot|_p^\alpha$ for some prime number p and $\alpha \in (0, \infty]$: $\mathcal{H}(x) \cong \mathbb{F}_p$ and the induced homomorphism $\tilde{\mathbb{Z}} = \mathbb{Z} \rightarrow \mathcal{H}(x) = \mathbb{F}_p$ is the natural projection $\mathbb{Z} \rightarrow \mathbb{F}_p$;
- $x = |\cdot|_0$: $\mathcal{H}(x) \cong \mathbb{Q}$ and the induced homomorphism $\tilde{\mathbb{Z}} = \mathbb{Z} \rightarrow \mathcal{H}(x) = \mathbb{Q}$ is the natural inclusion $\mathbb{Z} \rightarrow \mathbb{Q}$.

The following diagram illustrates the reduction map and the kernel map for the spectrum $\mathcal{M}(\mathbb{Z}, |\cdot|_0)$:



Proposition 1.23. Let R be a non-archimedean Banach ring and $\tilde{U} \subset \text{Spec}(\tilde{R})$ be a Zariski open subset. Then the preimage $\text{Red}^{-1}(\tilde{U})$ is a closed subset of $\mathcal{M}(R)$.

Proof. Yang: To be completed. □

Proposition 1.24. Let $\mathbf{k}$ be a complete non-archimedean field and A, B a strictly $\mathbf{k}$ -affinoid algebras. Suppose that $\varphi : A \rightarrow B$ is a bounded $\mathbf{k}$ -algebra homomorphism. Then we have:

- (a) if φ is finite, then the induced homomorphism $\tilde{\varphi} : \tilde{A} \rightarrow \tilde{B}$ is finite;
- (b) if φ is of finite type, then the induced homomorphism $\tilde{\varphi} : \tilde{A} \rightarrow \tilde{B}$ is of finite type.

Yang: To be revised.

1.3 Spectrum of Tate algebras

In this subsection, fix an algebraically closed complete non-archimedean field $\mathbf{k}$.

Spectrum of Tate algebra in one variable Let $A = \mathbf{k}\{T/r\}$. We list some types of points in the spectrum $\mathcal{M}(A)$.

Construction 1.25. For each $a \in \mathbf{k}$ with $|a| \leq r$, we have the *type I* point x_a corresponding to the evaluation at a , i.e., $|f|_{x_a} := |f(a)|$ for each $f \in A$.

For each closed disk $E = E(a, s) := \{b \in \mathbf{k} : |b - a| \leq s\}$ with center $a \in \mathbf{k}$ and radius $s \leq r$, we have the point $x_E = x_{a,s}$ corresponding to the multiplicative semi-norm defined by

$$|f|_{x_E} = |f|_{x_{a,s}} := \sup_{b \in E(a,s)} |f(b)|$$

for each $f \in A$. If $s \in |\mathbf{k}^\times|$, then the point x_E is called a *type II* point; otherwise, it is called a *type III* point.

Let $E_n = E(a_n, s_n)$ be a sequence of closed disks in $\mathbf{k}$ such that $E_{n+1} \subsetneq E_n$ and $\bigcap_n E_n = \emptyset$. Then we have the point $x_{\{E_n\}} = x_{\{a_n, s_n\}}$ corresponding to the multiplicative semi-norm defined by

$$|f|_{x_{\{E_n\}}} = |f|_{x_{\{a_n, s_n\}}} := \inf_n |f|_{x_{E_n}}$$

for each $f \in A$. Such a point is called a *type IV* point.

Proposition 1.26. The points in the spectrum $\mathcal{M}(\mathbf{k}\{r^{-1}T\})$ can be classified into four types as described in [Construction 1.25](#).

Proof. Fix $x \in \mathcal{M}(\mathbf{k}\{r^{-1}T\})$, set

$$s = \inf_{a \in \mathbf{k}} |T - a|_x \leq r, \quad E = \{a \in \mathbf{k} : |T - a|_x = s\} \subset E(0, r).$$

Case 1. $E \neq \emptyset$ and $s = 0$.

By assumption, there exists $a \in E$ such that $|T - a|_x = 0$. Note that if $f(a) = 0$, then $T - a \mid f$ in $\mathbf{k}\{r^{-1}T\}$ and hence $|f|_x = |T - a|_x |g|_x = 0$. Then we have

$$|f(a)| = ||f(a)| - |f(a) - f|_x| \leq |f|_x \leq |f(a)| + |f(a) - f|_x = |f(a)|$$

for each $f \in \mathbf{k}\{r^{-1}T\}$, which implies that $|f|_x = |f(a)|$. Thus x is a type I point x_a .

Case 2. $E \neq \emptyset$ and $s > 0$.

Let $a \in E$. Note that for every $b \in \mathbb{k}$, we have

$$|a - b| \leq \max\{|T - a|_x, |T - b|_x\} = |T - b|_x.$$

First we show that $E = E(a, s)$. For each $b \in E(a, s)$, we have $|T - b|_x \leq \max\{|T - a|_x, |a - b|\} = s$, which implies that $b \in E$. Conversely, for each $b \in E$, we have $|a - b| \leq \max\{|T - a|_x, |T - b|_x\} = s$.

Let $f \in \mathbb{k}[T]$ be a polynomial. Write $f = \prod_{i=1}^n (T - c_i)$ for some $c_1, \dots, c_n \in \mathbb{k}$. Then we have

$$|f|_x = \prod_{i=1}^n |T - c_i|_x \geq \prod_{i=1}^n |b - c_i| = |f(b)|, \quad \forall b \in E.$$

I claim that for every $\varepsilon \in (0, 1)$, there exists $b \in E$ such that $\varepsilon|T - c_i|_x < |b - c_i|$ for each $i = 1, \dots, n$. Indeed, if $c_i \notin E$, then $|T - c_i|_x = |b - c_i|$ for each $b \in E$. Hence we only need to consider the case when $c_i \in E$. Since $\mathbb{k}$ is algebraically closed, $E(a, s) \setminus \bigcup_{i=1}^n E(c_i, \varepsilon s) \neq \emptyset$. Choose b in the set. Then we have

$$|f(b)| \geq \prod_{i=1}^n \varepsilon |T - c_i|_x = \varepsilon^n |f|_x.$$

Thus $|f|_x = \sup_{b \in E} |f(b)|$ for each polynomial $f \in \mathbb{k}[T]$. Since polynomials are dense in $\mathbb{k}\{r^{-1}T\}$, we have $|f|_x = \sup_{b \in E} |f(b)|$ for each $f \in \mathbb{k}\{r^{-1}T\}$. Therefore, x is the point $x_E = x_{a,s}$, which is of type II or type III depending on whether $s \in |\mathbb{k}^\times|$ or not.

Case 3. $E = \emptyset$.

Set $E_n = \{a \in \mathbb{k} : |T - a|_x \leq s + 1/n\}$ and $a_n \in E_n$ for each $n \in \mathbb{N}$. By the similar argument as in [Case 2](#), we have $E_n = E(a_n, s + 1/n)$. Note that E_n is a decreasing sequence of closed disks with $\bigcap_n E_n = E = \emptyset$.

For $c \in \mathbb{k}$, there exists N such that $\forall n \geq N$, we have

$$c \notin E_n \implies |T - c|_x > |T - a_n|_x \implies |T - c|_x = |a_n - c|.$$

Thus

$$\inf_n |T - c|_{E_n} = \inf_n |a_n - c| = |T - c|_x.$$

By multiplicativity, we have $\inf_n |f|_{E_n} = |f|_x$ for each polynomial $f \in \mathbb{k}[T]$. And then by density of polynomials, the equality holds for each $f \in \mathbb{k}\{r^{-1}T\}$. Therefore, $x = x_{\{E_n\}} = x_{\{a_n, s_n\}}$ is of type IV. $\square$

Proposition 1.27. The completed residue fields of the four types of points in the spectrum $\mathcal{M}(\mathbb{k}\{r^{-1}T\})$ are described as follows:

- type I point x_a : $\mathcal{H}(x_a)$ is isomorphic to $\mathbb{k}$;
- type II point $x_{a,s}$: $\mathcal{H}(x_{a,s}) \cong \mathbb{k}_k((t))$;
- type III point $x_{a,s}$: $\mathcal{H}(x_{a,s}) \cong \mathbb{k}_k$ and the value group $|\mathcal{H}(x_{a,s})^\times|$ is generated by $|\mathbb{k}^\times|$ and s ;
- type IV point $x_{\{a_n, s_n\}}$: $\mathcal{H}(x_{\{a_n, s_n\}})$ is an immediate extension of $\mathbb{k}$.

Yang: To be checked.

Proof. **Yang:** To be completed. □

Example 1.28. The completed residue field $\mathcal{H}(x_a)$ for a type I point x_a with $a \in \mathbb{k}$ and $|a| \leq r$ is isomorphic to $\mathbb{k}$. **Yang:** To be complete.

Example 1.29. Here we describe the reduction map and the kernel map for the spectrum $\mathcal{M}(A)$. The reduction ring is

$$\tilde{A} = \begin{cases} \mathcal{K}_{\mathbb{k}}[T], & \text{if } r \in |\mathbb{k}^\times|; \\ \mathcal{K}_{\mathbb{k}}, & \text{if } r \notin |\mathbb{k}^\times|. \end{cases}$$

If $r \notin |\mathbb{k}^\times|$, then the reduction map $\text{Red} : \mathcal{M}(A) \rightarrow \text{Spec}(\tilde{A}) = \text{Spec}(\mathcal{K}_{\mathbb{k}})$ is trivial, i.e., every point in $\mathcal{M}(A)$ is mapped to the zero ideal of $\mathcal{K}_{\mathbb{k}}$. Hence we only consider the case when $r \in |\mathbb{k}^\times|$. Without loss of generality, we may assume that $r = 1$.

For each type I point x_a with $a \in \mathbb{k}$ and $|a| \leq 1$, the induced homomorphism $\tilde{A} = \mathcal{K}_{\mathbb{k}}[T] \rightarrow \mathcal{K}_{\mathcal{H}(x_a)} = \mathcal{K}_{\mathbb{k}}$ is given by $T \mapsto a \pmod{\mathbb{k}^\circ}$. Thus the reduction map $\text{Red} : \mathcal{M}(A) \rightarrow \text{Spec}(\tilde{A}) = \text{Spec}(\mathcal{K}_{\mathbb{k}}[T])$ sends the type I point x_a to the maximal ideal $(T - \tilde{a})$. For a type II or type III point $x_{a,s}$ with $a \in \mathbb{k}$ and $s \leq 1$, the homomorphism $A^\circ \rightarrow \mathcal{H}(x_{a,s})^\circ$

Yang: To be continued.

Example 1.30. Here we describe the kernel map for the spectrum $\mathcal{M}(A)$. If $x = x_a$ is a type I point corresponding to $a \in \mathbb{k}$ with $|a| \leq r$, then $\wp_{x_a} = (T - a)$. For type II, type III or type IV points x , note that if $f \in A$ is invertible, then $|f|_x > 0$ for each $x \in \mathcal{M}(A)$ since $|f|_x |f^{-1}|_x = |1| = 1$. By **Yang:** ref, we only need to consider elements in A of the form $T - c$ for some $c \in \mathbb{k}$ with $|c| \leq r$. Since $|T - c|_x \neq 0$ for all $c \in \mathbb{k}$ with $|c| \leq r$, we have $\wp_x = (0)$. To summarize, the kernel map $\mathcal{M}(A) \rightarrow \text{Spec}(A)$ is given by

$$\wp_x = \begin{cases} (T - a), & \text{if } x = x_a \text{ is a type I point;} \\ (0), & \text{if } x \text{ is a type II, type III or type IV point.} \end{cases}$$

Spectrum of Tate algebra in several variables Let $A = \mathbb{k}\{T\}$. To consider the spectrum $\mathcal{M}(A)$, inspired by [Example 1.30](#), we first look at the image of the points in $\mathcal{M}(A)$ under the kernel map $\mathcal{M}(A) \rightarrow \text{Spec}(A)$.

For every $\mathfrak{p} \in \text{Spec}(A)$, since A is noetherian, $\mathfrak{p}$ is finitely generated, say $\mathfrak{p} = (f_1, \dots, f_n)$. On $E(0, 1) \subset \mathbb{k}^n$, $\mathfrak{p}$ defines a closed analytic subspace $V = V(f_1, \dots, f_n) = \{x \in E(0, 1) : f_i(x) = 0, i = 1, \dots, n\}$. By **Yang:** ref, V is nonempty. On $E(0, 1)$, we have the supremum norm $\|\cdot\|_{\text{sup}}$. It inherits the metric from $E(0, 1)$ and hence is a complete metric space.

According to the proof of [Proposition 1.26](#), we should use prime divisor on $\text{Spec}(A/\mathfrak{p})$ to replace the role of points in one variable case.

Yang: To be continued.

2 Affinoid spaces

Fix a non-archimedean completed valued field $(\mathbf{k}, |\cdot|)$ throughout this section. Let A be a $\mathbf{k}$ -affinoid algebra, and let $X = \mathcal{M}(A)$ be the associated affinoid space. It should be viewed as the non-archimedean analogue of an affine variety in algebraic geometry or an affine scheme in scheme theory. The goal of

this section is to define a suitable Grothendieck topology on X such that we can define a structure sheaf of analytic functions on X , and then study its basic properties.

2.1 Affinoid domains

On a general open subset $U \subseteq X$, the ring of analytic functions on U may behave badly.

Yang: To be continued.

Definition 2.1. A closed subset $V \subseteq X$ is called an *affinoid domain* if there exists a $\mathbf{k}$ -affinoid algebra A_V and a morphism of $\mathbf{k}$ -affinoid algebras $\theta : A \rightarrow A_V$ satisfying the following universal property: for every bounded homomorphism of $\mathbf{k}$ -affinoid algebras $\psi : A \rightarrow B$ such that the induced map on spectra $\mathcal{M}(\psi) : \mathcal{M}(B) \rightarrow X$ has its image contained in V , there exists a unique bounded homomorphism $\varphi : A_V \rightarrow B$ such that the following right diagram commutes:

$$\begin{array}{ccc} & V & \\ \swarrow & & \nwarrow \\ X & \xleftarrow{\mathcal{M}(\psi)} & \mathcal{M}(B) \end{array} \quad \rightsquigarrow \quad \begin{array}{ccc} & A_V & \\ \nearrow \theta & & \nwarrow \varphi \\ A & \xrightarrow{\psi} & B \end{array}$$

In this case, we say that V is represented by the $\mathbf{k}$ -affinoid algebra A_V . If A_V is strict, then we say that V is a strict affinoid domain.

Slogan A closed subset $V \subset X$ is an affinoid domain if the functor “ $\text{Mor}(-, V)$ ” is representable by a $\mathbf{k}$ -affinoid algebra A_V .

There are several important classes of affinoid domains.

Construction 2.2. Let $f = (f_1, \dots, f_n), g = (g_1, \dots, g_m)$ be two tuples of elements in A and $r = (r_1, \dots, r_n), s = (s_1, \dots, s_m)$ be two tuples of positive real numbers. Consider the following closed subset of X :

$$X(\underline{f/r; s/g}) := \{x \in X : |f_i(x)| \leq r_i, |g_j(x)| \geq s_j, 1 \leq i \leq n, 1 \leq j \leq m\}.$$

Such a closed subset is called a *Laurent domain* of X . If $m = 0$, then it is called a *Weierstrass domain* of X , denoted by

$$X(\underline{f/r}) := \{x \in X : |f_i(x)| \leq r_i, 1 \leq i \leq n\}.$$

More generally, let $f = (f_1, \dots, f_n), g$ be elements in A such that the ideal generated by them is the whole algebra A . Set $p = (p_1, \dots, p_n)$ be a tuple of positive real numbers. We define the following closed subset of X :

$$X(\underline{(f/g)/p}) := \{x \in X : |f_i(x)| \leq p_i |g(x)|, 1 \leq i \leq n\}.$$

Such a closed subset is called a *rational domain* of X .

Proposition 2.3. Let A be a $\mathbf{k}$ -affinoid algebra, and let $X = \mathcal{M}(A)$ be the associated affinoid space. Then any Weierstrass domain, Laurent domain, or rational domain of X is an affinoid domain. Moreover, if A is strict, then these affinoid domains are all strict. Yang: To be checked.

Proof. Yang: To be added. □

We have a sequence of inclusion:

$$\{\text{Affinoid domains}\} \supseteq \{\text{Rational domains}\} \supseteq \{\text{Laurent domains}\} \supseteq \{\text{Weierstrass domains}\}.$$

Yang: Why Laurent domain is rational domain?

Yang: Is these domain open?

Proposition 2.4. Let $V \subseteq X$ be an affinoid domain represented by the $\mathbf{k}$ -affinoid algebra A_V . Then the natural map on spectra $\mathcal{M}(\theta) : \mathcal{M}(A_V) \rightarrow X$ induced by the representing homomorphism $\theta : A \rightarrow A_V$ has its image equal to V , and induces a homeomorphism $\mathcal{M}(A_V) \cong V$.

Proof. Yang: To be added. □

Proposition 2.5. Let $V \subseteq X$ be an affinoid domain represented by the $\mathbf{k}$ -affinoid algebra A_V . Then the natural bounded homomorphism $\varphi : A \rightarrow A_V$ is flat.

Proof. Yang: To be added. □

Proposition 2.6. Let $f : Y = \mathcal{M}(B) \rightarrow X = \mathcal{M}(A)$ be a morphism of $\mathbf{k}$ -affinoid spaces induced by a bounded homomorphism of $\mathbf{k}$ -affinoid algebras $\phi : A \rightarrow B$. Let $V \subseteq X$ be an affinoid domain represented by the $\mathbf{k}$ -affinoid algebra A_V . Then the inverse image $f^{-1}(V) \subseteq Y$ is an affinoid domain represented by the $\mathbf{k}$ -affinoid algebra $B \hat{\otimes}_A A_V$. Yang: To be revised.

Proof. Yang: To be added. □

Proposition 2.7. Let $V_1, V_2 \subseteq X$ be two affinoid domains represented by the $\mathbf{k}$ -affinoid algebras A_{V_1}, A_{V_2} , respectively. Then the intersection $V_1 \cap V_2 \subseteq X$ is an affinoid domain represented by the $\mathbf{k}$ -affinoid algebra $A_{V_1} \hat{\otimes}_A A_{V_2}$. Yang: To be checked.

Proof. Yang: To be added. □

Proposition 2.8. Let $x \in X$ be a point. Then the collection of strict affinoid domains of X containing x forms a fundamental system of neighborhoods of x . Explicitly, for any open neighborhood $U \subseteq X$ of x , there exists a strict affinoid domain $V \subseteq X$ such that $x \in V \subseteq U$. Yang: To be checked.

Theorem 2.9 (Gerritzen-Grauert theorem). Let $V \subseteq X$ be an affinoid domain. Then there exist finitely many rational domains $V_1, \dots, V_m \subseteq X$ such that $V = V_1 \cup \dots \cup V_m$. Yang: To be checked.

Proof. Yang: To be added. □

Definition 2.10. Let $V \subseteq X$ be an affinoid domain represented by the $\mathbf{k}$ -affinoid algebra A_V . Suppose that $A_V \hat{\otimes} \mathbf{K}$ is strictly $\mathbf{K}$ -affinoid for some completed valued field extension $\mathbf{K}/\mathbf{k}$. The *dimension* of V is defined as the Krull dimension of the affinoid algebra $A_V \hat{\otimes} \mathbf{K}$:

$$\dim(V) := \dim_{\text{Krull}}(A_V \hat{\otimes} \mathbf{K}).$$

By ??, the definition is independent of the choice of $\mathbf{K}$.

2.2 The Grothendieck topology and structure sheaf

Theorem 2.11 (Acyclicity theorem). Let A be a $\mathbf{k}$ -affinoid algebra, and let $X = \mathcal{M}(A)$ be the associated affinoid space. Let $\{V_i\}_{i \in I}$ be a finite covering of X by affinoid domains. For each finite subset $J \subseteq I$, set $V_J := \bigcap_{j \in J} V_j$, which is an affinoid domain by [Proposition 2.7](#). Then the augmented Čech complex

$$0 \rightarrow A \rightarrow \prod_{i \in I} A_{V_i} \rightarrow \prod_{\substack{J \subseteq I \\ |J|=2}} A_{V_J} \rightarrow \prod_{\substack{J \subseteq I \\ |J|=3}} A_{V_J} \rightarrow \cdots$$

is exact. **Yang: To be checked.**

Definition 2.12. A *special domain* of X is a finite union of affinoid domains of X . Let $V \subseteq X$ be a special domain, say $V = V_1 \cup \cdots \cup V_m$ with each V_i an affinoid domain. We define

$$A_V := \ker \left(\prod_{i=1}^m A_{V_i} \rightarrow \prod_{1 \leq i < j \leq m} A_{V_i \cap V_j} \right).$$

Yang: To be checked.

Proposition 2.13. Let $V \subseteq X$ be a special domain, say $V = V_1 \cup \cdots \cup V_m$ with each V_i an affinoid domain. Then V is an affinoid domain if and only if the $\mathbf{k}$ -algebra A_V defined in [Definition 2.12](#) is a $\mathbf{k}$ -affinoid algebra. In this case, V is represented by the $\mathbf{k}$ -affinoid algebra A_V . **Yang: To be added.**

Definition 2.14. The *site of special domains* on X , is defined as follows:

- the underlying category has as objects the special domains $V \subseteq X$, and the morphisms are given by the inclusion maps;
- a covering of an objects $V \subseteq X$ is a finite family of affinoid domains $\{V_i\}_{i \in I}$ such that $V = \bigcup_{i \in I} V_i$.

The *structure sheaf on site of special domains* is given by assigning to each special domain $V \subseteq X$ the ring A_V . **Yang: To be revised.**

Definition 2.15. For every open subset $U \subseteq X$, we define the ring of analytic functions on U as follows:

$$\mathcal{O}_X(U) := \varprojlim_{V \subseteq U} A_V \in \mathbf{Alg}_{\mathbf{k}},$$

where the limit is taken over all special domains $V \subseteq U$, and A_V is the $\mathbf{k}$ -algebra representing the special domain V . **Yang: To be revised.**

Remark 2.16. Note that in [Definition 2.14](#) we have defined a site on X consisting of special domains, and in [Definition 2.15](#) we have defined a structure sheaf on the usual topological space X . In fact, these two definitions are closely related. **Yang: To be revised.**

Definition 2.17. The *category of $\mathbf{k}$ -affinoid spaces* is defined as the opposite category of the category

of $\mathbf{k}$ -affinoid algebras:

$$\mathbf{k} := (\mathbf{Alg}_{\mathbf{k}}^{\text{aff}})^{\text{op}}.$$

Yang: To be revised.

Yang: Coherent sheaf.

2.3 Boundary and interior

Let A, B be two $\mathbf{k}$ -affinoid algebras, and let $f : Y = \mathcal{M}(B) \rightarrow X = \mathcal{M}(A)$ be a morphism of $\mathbf{k}$ -affinoid spaces induced by a bounded homomorphism of $\mathbf{k}$ -affinoid algebras $\phi : A \rightarrow B$.

Definition 2.18. The *relative interior* of Y over X is defined as

$$(Y/X) := \{y \in Y : \text{there exists an admissible surjective } A \rightarrow B \text{ inducing } y\}.$$

Yang: To be revised.

Definition 2.19.

2.4 Some local properties

Yang: The local ring $\mathcal{O}_{X,x}$.

Yang: The residue field and its completion.

Example 2.20. Yang: To be added.

Proposition 2.21. The local ring $\mathcal{O}_{X,x}$ at any point $x \in X$ is noetherian.

Definition 2.22. We say that the affinoid space X has a local property $\mathcal{P}$ at a point $x \in X$ if the local ring $\mathcal{O}_{X,x}$ has the property $\mathcal{P}$. Yang: To be added.

3 Analytic spaces

Fix a complete non-archimedean field $\mathbf{k}$.

3.1 Definition of analytic spaces

Definition 3.1. Let X be a topological space. A *net* (resp. *quasi-net*) on X is a collection τ of subsets of X such that

- (a) for every $x \in X$, there exist $U_1, \dots, U_n \in \tau$ such that $x \in U_1 \cap \dots \cap U_n$ and $U_1 \cup \dots \cup U_n$ is a neighborhood of x in X ;
- (b) for every $U, V \in \tau$, the collection of all $W \in \tau$ such that $W \subseteq U \cap V$ is a (resp. quasi-) net on $U \cap V$.

Yang: To be checked.

Definition 3.2. A $\mathbf{k}$ -*quasiaffinoid space* is a locally ringed space $(X, \mathcal{O}_X)$ which is isomorphic to an open subspace of a $\mathbf{k}$ -affinoid space.

Definition 3.3. A $\mathbf{k}$ -*analytic space* is a locally ringed space $(X, \mathcal{O}_X)$ such that every point $x \in X$ has an open neighborhood U such that $(U, \mathcal{O}_X|_U)$ is a $\mathbf{k}$ -quasiaffinoid space.

Remark 3.4. There are two different versions of the definition of $\mathbf{k}$ -analytic spaces in the literature: one given by Berkovich in

Definition 3.5. Let $(X, \mathcal{O}_X)$ and $(Y, \mathcal{O}_Y)$ be $\mathbf{k}$ -analytic spaces. A *morphism* $f : (X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y)$ of $\mathbf{k}$ -analytic spaces is a morphism of locally ringed spaces. The category of $\mathbf{k}$ -analytic spaces is the category whose objects are $\mathbf{k}$ -analytic spaces and whose morphisms are morphisms of $\mathbf{k}$ -analytic spaces. Yang: To be revised.

Theorem 3.6. Let $(X, \mathcal{O}_X)$, $(Y, \mathcal{O}_Y)$, and $(Z, \mathcal{O}_Z)$ be $\mathbf{k}$ -analytic spaces, and let $f : (X, \mathcal{O}_X) \rightarrow (Z, \mathcal{O}_Z)$ and $g : (Y, \mathcal{O}_Y) \rightarrow (Z, \mathcal{O}_Z)$ be morphisms of $\mathbf{k}$ -analytic spaces. Then the fibre product $(X, \mathcal{O}_X) \times_{(Z, \mathcal{O}_Z)} (Y, \mathcal{O}_Y)$ exists in the category of $\mathbf{k}$ -analytic spaces. Yang: To be revised.

Definition 3.7. Let $(X, \mathcal{O}_X)$ be a $\mathbf{k}$ -analytic space. We say that $(X, \mathcal{O}_X)$ is *good* if every point $x \in X$ has an open neighborhood U such that $(U, \mathcal{O}_X|_U)$ is a $\mathbf{k}$ -affinoid space.

3.2 The first properties

Definition 3.8. Let $f : (X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y)$ be a morphism of $\mathbf{k}$ -analytic spaces. We say that f is *finite* if for every affinoid domain $V \subseteq Y$, the preimage $f^{-1}(V)$ is an affinoid domain in X and the induced morphism $\mathcal{O}_Y(V) \rightarrow \mathcal{O}_X(f^{-1}(V))$ is a finite morphism of $\mathbf{k}$ -affinoid algebras. Yang:

Theorem 3.9. Let $f : (X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y)$ be a morphism of $\mathbf{k}$ -analytic spaces. Then there exists a factorization $f = g \circ h$ where $h : (X, \mathcal{O}_X) \rightarrow (Z, \mathcal{O}_Z)$ is a finite morphism and $g : (Z, \mathcal{O}_Z) \rightarrow (Y, \mathcal{O}_Y)$ is a morphism such that for every affinoid domain $W \subseteq Y$, the preimage $g^{-1}(W)$ is an affinoid domain in Z and the induced morphism $\mathcal{O}_Y(W) \rightarrow \mathcal{O}_Z(g^{-1}(W))$ is a flat morphism of $\mathbf{k}$ -affinoid algebras.

Yang:

Definition 3.10. Let $f : (X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y)$ be a morphism of $\mathbf{k}$ -analytic spaces. We say that f is *separated* if the diagonal morphism $\Delta_f : (X, \mathcal{O}_X) \rightarrow (X, \mathcal{O}_X) \times_{(Y, \mathcal{O}_Y)} (X, \mathcal{O}_X)$ is a closed immersion.

Yang:

Definition 3.11. Let $f : (X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y)$ be a morphism of $\mathbf{k}$ -analytic spaces. We say that f is *proper* if it is separated, compact, and without boundary. Yang:

Theorem 3.12. Let $f : (X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y)$ be a proper morphism of $\mathbf{k}$ -analytic spaces. Then for every coherent $\mathcal{O}_X$ -module $\mathcal{F}$, the direct image $f_*(\mathcal{F})$ is a coherent $\mathcal{O}_Y$ -module. Yang:

Definition 3.13. Let $f : (X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y)$ be a morphism of $\mathbf{k}$ -analytic spaces. We say that f is *smooth* (resp. *étale*) if for every $x \in X$, there exist open neighborhoods U of x in X and V of $f(x)$ in Y such that $f(U) \subseteq V$ and a commutative diagram Yang:

4 GAGA

In this section, we discuss the GAGA (Géométrie Algébrique et Géométrie Analytique) principle for Berkovich spaces.

Construction 4.1. Let X be a scheme of finite type over a non-archimedean field k . We can associate to X its Berkovich analytification X^{an} , which is a Berkovich analytic space over k .

The analytification functor $(\cdot)^{\text{an}}$ has the following properties:

- It is compatible with fiber products: for schemes X and Y over k , we have $(X \times_k Y)^{\text{an}} \cong X^{\text{an}} \times_k Y^{\text{an}}$.
- It preserves open and closed immersions.
- It induces a homeomorphism between the underlying topological spaces of X and X^{an} .
- For a coherent sheaf $\mathcal{F}$ on X , there is an associated coherent sheaf $\mathcal{F}^{\text{an}}$ on X^{an} .

Yang: To be revised.